Noa Chang

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Experience

Software Engineer Intern, Backend – Crypto Core / Crypto Trading May 2026 – Aug. 2026
Robinhood *Toronto, ON*

- Designed and shipped an end-to-end crypto tax-withholding reversal pipeline for a regulatory deadline, spanning bulk ingestion, validation, asynchronous processing, reversal/rebooking, accounting events, and operational tooling.
- Built idempotent money-movement workflows across sharded services using uniqueness constraints, execution-time validation, row-level locking, and append-only audit events to prevent duplicate reversals under retries and re-uploaded batches.
- Shipped 3 production workflows for Tax Operations, including bulk reversal APIs, asynchronous processing and status tracking, and withholding-history auditing; incorporated operator feedback through hands-on demos and production validation.
- Delivered 63 merged changes across 10+ services and components, additionally shipping Crypto IRA eligibility restrictions, statement tooling, and production-safety improvements; reviewed 50+ teammate PRs.

Machine Learning Engineer Intern, AI Team May 2025 – Aug. 2025
SK AX, Enterprise AI Division of SK Group *Icheon, South Korea*

- Maintained and optimized CI/CD pipelines on strict on-premise infrastructure using Jenkins, ensuring secure processing of TB-scale manufacturing logs; integrated Pylint and unit tests, reducing build failure rates by 20%.
- Developed Python resource-profiling scripts with psutil to monitor closed-network inference services, identifying memory leaks that reduced latency by 15% and eliminated weekly OOM crashes.
- Engineered schema validation and null-detection checks within ETL workflows, preventing corrupted data ingestion and improving downstream model-training reliability.

Research Intern Sep. 2024 – Dec. 2024
Ontario Nutri Lab *Fergus, ON*

- Consolidated 50+ trial logs into a unified SQL database using Python and Pandas, reducing retrieval time from hours to seconds and enabling reproducible analysis across experiments.
- Cross-validated Random Forest and XGBoost models to identify key response drivers; addressed class imbalance with SMOTE and achieved an F1-score of 0.85.

Data Analyst Jan. 2022 – Jul. 2022
Yes Pharm *Seoul, South Korea*

- Restructured analytics data into a Star Schema, optimizing query performance by 91%, from 12 seconds to 1 second, and automated daily reporting workflows with Python and SQL.

Projects

Provenance-Preserving Media Archive Nov. 2026 – Present
Python, FastAPI, Pydantic, SQLite, TypeScript, Next.js, React, GitHub Actions

- Built a multi-source ingestion pipeline consolidating 13K+ evidence items from 6 platforms into 5.3K normalized records, with resumable idempotent imports, per-revision content hashing, and a SHA-256 content-addressed media store.
- Designed a review ledger that binds each of 1.4K reviewer decisions to its record hash and rejects stale facts, feeding hash-manifested release candidates gated by Pydantic/JSON Schema export contracts.
- Developing the Next.js front end (cursor-paginated search, calendar/iCal feed, fail-closed publishing); 570+ automated tests across both codebases with GitHub Actions CI.

GuideU Apr. 2023 – Mar. 2024
Python, PyTorch, Whisper, YOLOv8, Pandas

- Built and instrumented real-time speech-to-text and computer-vision pipelines, analyzing latency and failure modes to evaluate model and system performance.

WakAI Jan. 2024 – Dec. 2024
Python, SQL, Prompt Engineering, PII Redaction

- Designed prompt and data workflows for 40+ persona chatbots, implementing SQL-backed pipelines and PII redaction for safer user-facing AI interactions.

Technical Skills

Languages: Python, Go, C/C++, JavaScript/TypeScript, C#, SQL/Postgres, R, HTML/CSS
Backend/Systems: Django, Flask, FastAPI, REST APIs, async processing, distributed systems, testing, CI/CD, data integrity
Developer Tools: Git, Graphite, Linux, Docker, Kubernetes, Jenkins, GitHub Actions, Bash
Data/ML: Pandas, NumPy, scikit-learn, PyTorch, XGBoost, YOLOv8, ETL workflows, model evaluation

Education

University of Waterloo Waterloo, ON
Bachelor of Mathematics, Applied Mathematics *Expected Graduation: Dec. 2027*